

```
def find_s(training_data):  
    # Student Details  
    name = "Shivani Kumari"  
    reg_no = "24BECS105"  
  
    # Initialize the specific hypothesis with the first positive instance  
    hypothesis = None  
  
    for row in training_data:  
        if row[-1] == 'Yes':  
            # Target concept is positive  
            hypothesis = row[:-1].copy()  
            break  
  
    if hypothesis is None:  
        return "No positive instances found."  
  
    # Update hypothesis based on remaining positive instances  
    for row in training_data:  
        if row[-1] == 'Yes':  
            for i in range(len(hypothesis)):  
                if row[i] != hypothesis[i]:  
                    hypothesis[i] = '?'  
  
    return hypothesis
```

```
def find_s(training_data):
```

```
    return hypothesis
```

```
# Example Usage
```

```
if __name__ == "__main__":
```

```
    # Student Details
```

```
    name = "Shivani Kumari"
```

```
    reg_no = "24BECS105"
```

```
    # Columns: Sky, Temp, Humidity, Wind, Water, Forecast, EnjoySport
```

```
    dataset = [
```

```
        ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
```

```
        ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
```

```
        ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
```

```
        ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
```

```
    ]
```

```
    most_specific_hypothesis = find_s(dataset)
```

```
    # Display Student Details
```

```
    print("=====")
```

```
    print("Name      :", name)
```

```
    print("Reg No.   :", reg_no)
```

```
# Columns: Sky, Temp, Humidity, Wind, Water, Forecast, EnjoySport
dataset = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]
```

```
most_specific_hypothesis = find_s(dataset)
```

```
# Display Student Details
```

```
print("=====")
print("Name    :", name)
print("Reg No :", reg_no)
print("=====")
```

```
print(
    "Most Specific Hypothesis found by Find-S:"
```

Output



=====

Name : Shivani Kumari

Reg No : 24BECS105

=====

Most Specific Hypothesis found by Find-S: ['Sunny', 'Warm', '?', 'Strong', '?', '?']

=== Code Execution Successful ===

